

DEPARTMENT OF

CIVIL

ENGINEERING

VISION OF THE DEPARTMENT

To develop Civil Engineering Department as a Centre of excellence for imparting value based education to the students at undergraduate and post-graduate level to meet industry needs and to develop as a major research center meeting national and international standards.

MISSION OF THE DEPARTMENT

- To impart in-depth and up-to-date knowledge of Civil Engineering concepts with focus on character enhancement, leadership qualities, effective communication, social responsibility and pursuit of lifelong learning and professional development.
- To provide a platform to the students to engage in original innovative research.

**B. TECH.
(CIVIL ENGINEERING)**

B. TECH. (CE)

PROGRAM EDUCATIONAL OBJECTIVES

PEO-I: To provide students with a solid foundation in Basic and Engineering Sciences to understand, analyze and evaluate the knowledge concept to achieve expertise in core areas of Civil Engineering.

PEO-II: To prepare the students to achieve high level technical expertise in the fields of Civil Engineering and to excel in the design and construction of various components or systems of Civil Engineering and to make the students capable of pursuing higher studies and research.

PEO-III: To establish acquaintance with the practical implementation of the theoretical concepts through laboratories, by bringing the real world into the academics through virtual industry labs and to enhance experimental skills of students beyond curriculum and encouraging them to identify and solve complex engineering problems.

PEO-IV: To equip students with modern professional abilities such as effective communication, collaborative work in diverse teams, ethical decision making, successful management of personal and professional career objectives and passion for continuous development through lifelong learning.

PEO-V: To prepare the students to guide their professional development by bringing awareness of professional society activities, professional licensure requirements and opportunities for further education in graduate school.

B. TECH. (CE)

PROGRAM OUTCOMES

PO-1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO-2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).

PO-3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).

PO-4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO-5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including

prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).

PO-6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO-7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).

PO-8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO-9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO-10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO-11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii)

adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

B. TECH. (CE)

PROGRAM SPECIFIC OUTCOMES

PSO-1: Graduates will be able to apply, analyze, and design solutions in the domains of Structural, Geotechnical, Transportation, Environmental, and Water Resources Engineering to address real-world civil engineering problems.

PSO-2: Graduates will be able to demonstrate proficiency in using modern engineering tools, software, and laboratory techniques for the analysis, design, and management of civil engineering projects.

PSO-3: Graduates will be able to plan, design, and implement sustainable infrastructure systems considering environmental, societal, and ethical aspects, thereby contributing to Sustainable Development Goals (SDGs) and professional practices.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD

B. TECH. I YEAR
CIVIL ENGINEERING

I SEMESTER

R25

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT101	Matrices and Calculus	3	1	0	4	4
25BS1PH101	Applied Physics	3	0	0	3	3
25HS1EN101	English for Skill Enhancement	3	0	0	3	3
25PC1CE101	Engineering Geology and Building Materials	4	0	0	4	4
25ES1EE104	Elements of Electrical and Electronics Engineering	2	0	0	2	2
25BS2PH101	Applied Physics Laboratory	0	0	2	2	1
25HS2EN101	English Language and Communication Skills Laboratory	0	0	2	2	1
25PC2CE101	Engineering Geology and Building Materials Testing Laboratory	0	0	2	2	1
25ES2ME101	Engineering Workshop	0	0	2	2	1
25MN6HS101	Induction Programme	2	0	0	2	0
Total		17	1	8	26	20

II SEMESTER

R25

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT105	Ordinary Differential Equations and Vector Calculus	2	1	0	3	3
25BS1CH102	Engineering Chemistry	3	0	0	3	3
25PC1CE102	Surveying	3	0	0	3	3
25ES1CE101	Applied Mechanics	2	1	0	3	3
25PC1CE103	Building Planning and Construction	3	0	0	3	3
25ES3CE101	Computer Aided Engineering Graphics	0	0	6	6	3
25BS2CH101	Engineering Chemistry Laboratory	0	0	2	2	1
25PC2CE102	Surveying Laboratory	0	0	2	2	1
25MN6HS102	Environmental Science	2	0	0	2	0
Total		15	2	10	27	20

L – Lecture T – Tutorial P – Practical D – Drawing
 C – Credits SE – Sessional Examination CA – Class Assessment
 SEE – Semester End Examination D-D – Day to Day Evaluation
 CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
 ELA – Experiential Learning Assessment
 LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS1MT101) MATRICES AND CALCULUS

TEACHING SCHEME		
L	T/P	C
3	1	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematical Knowledge at Pre-University level

COURSE OBJECTIVES:

- To learn the concept of a rank of the matrix and applying this concept to check the consistency of a system of linear equations
- To learn the concept of eigen values and eigen vectors and to reduce the quadratic form to canonical form
- To know the geometrical approach to the mean value theorems and their application to the mathematical problems
- To find the maxima and minima of function of two and three variables
- To evaluate multiple integrals and their applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Compute the rank of a matrix and analyze the solution of a system of linear equations

CO-2: Calculate the Eigenvalues and Eigen vectors and Reduce the quadratic form to canonical form using orthogonal transformations

CO-3: Solve the applications on the mean value theorems

CO-4: Find the extreme values of functions of two and three variables with/ without constraints

CO-5: Evaluate the multiple integrals and apply the concept to find areas, volumes

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	1	1	1	1	-	2	1	2	2	1	-
CO-2	3	2	2	1	2	1	1	-	2	1	2	2	1	-
CO-3	3	2	2	1	1	1	1	-	2	1	2	2	1	-
CO-4	3	2	2	1	2	1	1	-	2	1	2	2	1	-
CO-5	3	2	2	1	2	1	1	-	2	1	2	2	1	-

UNIT-I:

Matrices: Rank of a matrix by Echelon form and Normal form – Inverse of Non-singular matrices by Gauss-Jordan method. System of linear equations: Solving system of Homogeneous and Non-Homogeneous equations, Gauss Seidel Iteration Methods.

UNIT-II:

Eigenvalues and Eigenvectors: Eigenvalues – Eigenvectors and their properties – Diagonalization of a matrix – Cayley-Hamilton Theorem (without proof) – finding inverse and power of a matrix by Cayley-Hamilton Theorem – Quadratic forms and

Nature of the Quadratic Forms – Reduction of Quadratic form to canonical forms by Orthogonal Transformation.

UNIT-III:

Single Variable Calculus: Limits and Continuous function and its properties. Mean value theorems: Rolle's theorem – Lagrange's Mean value theorem with their Geometrical Interpretation and applications – Cauchy's Mean value Theorem – Taylor's Series (All the theorems without proof).

Curve Tracing: Curve tracing in cartesian and polar coordinates

UNIT-IV:

Partial Differentiation and Its Applications: Definitions of Limit and continuity – Partial Differentiation: Euler's Theorem – Total derivative – Jacobian – Functional dependence & independence. Applications: Maxima and minima of functions of two variables and three variables using method of Lagrange multipliers.

UNIT-V:

Multiple Integrals: Evaluation of Double Integrals (Cartesian and polar coordinates) – change of order of integration (only Cartesian form) – Change of variables for double integrals (Cartesian to polar). Evaluation of Triple Integrals – Change of variables for triple integrals (Cartesian to Spherical and Cylindrical polar coordinates). Applications: Areas by double integrals and volumes by triple integrals.

TEXT BOOKS:

1. Higher Engineering Mathematics, B. S. Grewal, 36th Edition, Khanna Publishers, 2010
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, 5th Edition, Narosa Publications, 2016

REFERENCES:

1. Advanced Engineering Mathematics, Erwin Kreyszig, 13th Edition, John Wiley & Sons, 2006
2. Thomas' Calculus, G. B. Thomas, 13th Edition, Pearson, 2014
3. A text Book of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications, 2008
4. Higher Engineering Mathematics, H. K. Dass and Er. Rajnish Verma, S. Chand

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS1PH101) APPLIED PHYSICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: 10 + 2 Physics

COURSE OBJECTIVES:

- To explore the working and applications of lasers and fibre optics in modern technology
- To study crystal structures and defects in materials
- To learn nano materials and characterization techniques like XRD & SEM
- To understand the properties and applications of dielectric and magnetic materials
- To identify the importance of quantum mechanics and quantum computing

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Explain the principles of lasers and fibre optics and their applications in communication and sensing

CO-2: Understand different types of crystals and the importance of crystal defects

CO-3: Identify the role of nanomaterials in science and engineering applications

CO-4: Classify the dielectric and magnetic materials to illustrate their applications

CO-5: Realize the role of quantum mechanics in quantum computing

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	1	-	2	1	-	-	-	1	2	-	-
CO-2	3	3	-	-	-	-	1	-	-	-	1	1	-	-
CO-3	3	2	2	1	1	2	1	-	-	-	1	-	-	-
CO-4	3	3	1	1	1	2	1	-	-	-	1	1	-	-
CO-5	3	3	1	1	1	2	1	-	-	-	1	-	-	-

UNIT-I:

Laser and Fibre Optics: Introduction to laser, characteristics of laser, Einstein coefficients and their relations, metastable state, population inversion, pumping, lasing action, Ruby laser, He-Ne laser, Semiconductor laser, applications: Bar code scanner, LIDAR for autonomous vehicle.

Introduction to fibre optics, total internal reflection, construction of optical fibre, acceptance angle, numerical aperture, classification of optical fibres, losses in optical fibre, applications: optical fibre for communication system, sensor for structural health monitoring.

UNIT-II:

Crystallography & Defects: Introduction: Unit cell, space lattice, basis, lattice parameters; crystal structures, Bravais lattices, packing factor: SC, BCC, FCC; Miller indices, inter-planar distance; Introduction to defects, Point defects (Vacancies, Interstitial and Impurities) Line imperfections, Edge and Screw dislocation, Burger vector, Surface defects and volume defects (Qualitative Treatment).

UNIT-III:

Nano materials & Characterization: Concept of Nanomaterials: surface to volume ratio and quantum confinement, Fabrication: Sol-Gel, Ball milling, Applications. **Characterization:** X-ray diffraction, Bragg's law, powder method, Calculation of average crystallite size using Debye Scherrer's formula, scanning electron microscopy (SEM): block diagram, working principle.

UNIT-IV:

Dielectric and Magnetic Materials: Introduction to dielectric materials, types of polarization (qualitative): electronics, ionic & orientation; ferroelectric, piezoelectric, pyroelectric materials and their applications: Ferroelectric Random-Access Memory (Fe-RAM), load cell and fire sensor.

Introduction to magnetic materials, classification of magnetic materials, hysteresis, Weiss domain theory of ferromagnetism, soft and hard magnetic materials, applications: magnetic hyperthermia for cancer treatment, magnets for EV, Giant Magneto Resistance (GMR) device.

UNIT-V:

Quantum Mechanics and Quantum Computing: Introduction, de-Broglie hypothesis, Heisenberg uncertainty principle, physical significance of wave function, postulates of quantum mechanics: operators in quantum mechanics, eigen values and eigen functions, expectation value; Schrödinger's time independent wave equation, particle in a 1D box.

Introduction, concept of quantum computer, advantages of quantum computing over classical computation. classical bits, Qubits, multiple Qubit system, quantum computing system for information processing, Dirac's Bra and Ket notation and their properties, Introduction to entanglement.

TEXT BOOKS:

1. Textbook of Engineering Physics, M. N. Avadhanulu, P. G. Kshirsagar & T. V. S. Arun Murthy, 11th Edition, S. Chand Publications, 2019
2. Introduction to Solid State Physics, Charles Kittel, John Wiley & Sons
3. Introduction to Classical and Quantum Computing, Thomas G. Wong, Rooted Grove

REFERENCES:

1. Engineering Physics, B. K. Pandey and S. Chaturvedi, 2nd Edition, Cengage Learning, 2022
2. Energy Materials a Short Introduction to Functional Materials for Energy Conversion and Storage, A. S. Bandarenka, CRC Press, Taylor & Francis Group Energy Materials, Taylor & Francis Group, 1st Edition, 2022
3. Concepts of Modern Physics, Aurther Beiser, McGraw-Hill

4. Engineering Physics, P. K. Palanisamy, SciTech Publications
5. Quantum Computing, Jozef Gruska, McGraw-Hill

ONLINE RESOURCES:

1. <https://shijuinpallotti.wordpress.com/wp-content/uploads/2019/07/optical-fiber-communications-principles-and-pr.pdf>
2. https://www.geokniga.org/bookfiles/geokniga-crystallography_0.pdf
3. <https://dpbck.ac.in/wp-content/uploads/2022/10/Introduction-to-Solid-State-PhysicsCharles-Kittel.pdf>
4. <https://www.thomaswong.net/introduction-to-classical-and-quantum-computing-1e4p.pdf>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25HS1EN101) ENGLISH FOR SKILL ENHANCEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Basic English

COURSE OBJECTIVES:

- To enhance vocabulary through word formation processes
- To read and comprehend different kinds of texts
- To write clear, concise paragraphs, essays, and letters to produce appropriate technical prose
- To improve coherence and cohesion in writing and speaking
- To recognize and practice the use of rhetorical elements necessary for the successful practice of scientific and technical communication

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Use vocabulary contextually and effectively in oral and written communication

CO-2: Demonstrate an understanding of the rules of functional Remedial Grammar and sentence structures

CO-3: Demonstrate improved competence in Standard Written English, including Remedial Grammar, sentence and paragraph structure, and coherence, and use this knowledge to accurately communicate technical information

CO-4: Apply principles of writing while developing paragraphs, essays, letters and technical prose

CO-5: Employ appropriate rhetorical patterns of discourse in scientific and technical communication

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	1	1	1	1	-	2	1	1	2	3	2	-	1	1
CO-2	2	2	2	2	2	2	2	1	3	3	2	-	1	-
CO-3	2	2	2	2	1	3	2	1	3	3	2	-	1	1
CO-4	1	1	1	1	1	2	2	1	2	3	2	1	2	1
CO-5	1	1	1	1	1	2	1	1	2	2	1	2	2	-

UNIT-I:

1. **Reading:** -The Generation Gap' by Benjamin M. Spock
The Lotos Eaters – Alfred Tennyson, The Land of High Passes – Bhagyalakshmi Krishnamurthy
2. **Grammar:** Degrees of comparisons, Conjunctions, and Prepositions – Identifying Common Errors
3. **Vocabulary:** Word Formation (Affixation, Compounding, Conversion, Blending, Borrowing)

4. **Writing:** Punctuation, Clauses and Sentences, Transitional Devices, Paragraph Writing- Process

UNIT-II:

1. **Reading:** Emerging Technologies
Medicate – Oleksandr Gorpynich Of Studies – Francis Bacon
2. **Grammar:** Articles Noun-Pronoun Agreement; Concord
3. **Vocabulary:** Word Formation (Prefixes, Suffixes, Root Words)
4. **Writing:** Essay Writing -Descriptive, Argumentative, Expository

UNIT-III:

1. **Reading: Leisure- William Henry Davies -Be Thankful - Unknown Author**
The Rising of the Moon – Lady Gregory, With The Photographer – Stephen Leacock
2. **Grammar:** Misplaced Modifiers
3. **Vocabulary:** Synonyms and Antonyms
4. **Writing:** Letter Writing- Formal Letter Writing– Letter of Complaint, Letter of Requisition, Email Writing, Email Etiquette

UNIT-IV:

1. **Reading:** -Why a Start-Up Needs to Find its Customers First'- Pranav Jain
Too dear – Leo Tolstoy, Shooting an Elephant – George Orwell
2. **Grammar:** Clichés, Redundancies
3. **Vocabulary:** Common Abbreviations
4. **Writing:** Summary Writing, Job Application, Resume Writing

UNIT-V:

1. Reading: '**Professional Ethics**'
2. Patterns of Writing: Comparison and Contrast
3. Patterns of Writing: Cause and Effect Paragraphs
4. Patterns of Writing: Classification Paragraph
5. Patterns of Writing: Problem–Solution Pattern of Writing

TEXT BOOKS:

1. Board of Editors, English for the Young in the Digital World, Orient Black Swan, 2025
2. Unlocking English: A Skill-Based Approach for Language Proficiency, Cengage Publications, 2025
3. Technical Communication, Rebecca E. Burnett, 6th Edition, Cengage Learning

REFERENCES:

1. Practical English Usage, Swan Michael, New Edition, Oxford University Press, 2016
2. English Grammar Just for You, Karal, Rajeevan, Oxford University Press, 2023
3. Empowering with Language: Communicative English for Undergraduates, Cengage Learning, 2024
4. Communication Skills – A Workbook, Sanjay Kumar & Pushp Lata, Oxford University Press, 2022
5. Study Writing, Hamp Liz., Lyons and Heasley, Ben, C U P, 2006

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25PC1CE101) ENGINEERING GEOLOGY AND BUILDING MATERIALS

TEACHING SCHEME		
L	T/P	C
4	0	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To identify the various physical properties of minerals, rocks, and to understand the concept of structural geology
- To understand basic structural geology concepts from a civil engineering point of view
- To comprehend the characteristics of different building materials
- To acquire knowledge about the applications of building materials in civil engineering

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify the physical properties of minerals and rocks and explain basic concepts of structural geology

CO-2: Interpret and differentiate structural geological concepts from a civil engineering perspective

CO-3: Examine and evaluate the characteristics of different building materials

CO-4: Apply knowledge of building materials in civil engineering applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	-	-	-	-	1	-	-	1	-	1	3	-	-
CO-2	3	-	-	1	-	1	-	-	1	-	2	3	-	-
CO-3	3	-	-	-	-	1	-	-	1	-	2	3	-	-
CO-4	3	-	-	-	-	1	-	-	1	-	2	3	-	-

UNIT-I:

Introduction: Definition of Geology, Engineering Geology. Importance of geology from a civil engineering point of view.

Mineralogy: Definition of mineral, Importance of the study of minerals: rock-forming and ore-forming minerals. Study of minerals by the physical identification method and their properties - Physical properties of the rock forming minerals - Feldspar, Quartz, Augite, Olivine, Mica, Garnet, Physical properties of the ore forming minerals - Pyrite, Hematite, Magnetite, Galena, Graphite, and Bauxite.

UNIT-II:

Petrology: Definition of a rock - Classification of rocks - Geological classification of rocks - Rock cycle - Classification of forms - structures and textures of igneous rocks - Megascopic Study of Granite - Dolerite and Basalt - Classification of sedimentary rocks - its structures - and textures - Megascopic Study of Sandstone - Shale and Limestone - Classification of metamorphic rocks - their structures and textures - Megascopic Study of Gneiss - Quartzite and Marble.

UNIT-III:

Structural Geology: Study of geological structures associated with rocks such as Folds – Faults – Joints - Unconformities, and their important types

Building Stones: Properties- Structural requirements – Quarrying - Dressing of stones – Various Tests for stones

UNIT-IV:

Bricks, Cement, Wood:

Bricks: Composition of Brick earth – Manufacturing and Properties of Bricks - Tests on Bricks - Classification of Bricks – Types of Bricks: Fly Ash Bricks – Autoclaved Aerated Concrete (AAC) Bricks, Applications – Cement : Composition of cement and their functions, Types of cement, Wood : Structure – Properties – Seasoning Methods

UNIT-V:

Steel, Glass, Paints, and Modern Materials for Construction: **Steel:** Composition - Properties of Mild Steel & HYSD bars - Market forms of Steel - **Glass:** Composition – Applications - **Paints:** Types - Heat Insulating Materials - Recycled Aggregates - Smart Materials - Composite Materials- Types and Applications

TEXT BOOKS:

1. Textbook of Engineering Geology, N. Chenna Kesavulu, 3rd Edition, Laxmi Publications, 2018
2. Engineering and General Geology, Parbin Singh, 8th Edition, S. K. Kataria & Sons, 2023
3. Building Materials, S. K. Duggal, 6th Edition, New Age International Publishers , 2025

REFERENCES:

1. Engineering Materials S. C. Rangwala, 44th Edition, Charotar Publishing House , 2024
2. Fundamentals of Engineering Geology, F. G. Bell, BS Publications, 2020
3. Engineering Geology, D. Venkat Reddy, 2nd Edition, Vikas Publishing House, 2017
4. Building Materials Products, Properties and Systems, M. L. Gambhir and Neha Jamwal, 1st Edition, Tata McGraw-Hill Education, 2018
5. Building Materials, P. C. Varghese, 2nd Edition, PHI Learning, 2015

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES1EE104) ELEMENTS OF ELECTRICAL AND ELECTRONICS ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematics, Physics

COURSE OBJECTIVES:

- To understand the use of electrical energy in different engineering fields
- To analyze electrical circuits using different network reduction techniques
- To practice the techniques to control, assess electrical machines and installations
- To learn principles of operation, construction and characteristics of various electronic devices

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Appreciate the role of electrical energy in various engineering branches and analyze the DC circuits using various network reduction techniques

CO-2: Analyze the various electrical parameters of AC circuits with R-L-C elements

CO-3: Understand the operation of various Electrical Machines & Installations

CO-4: Apply basic electronics device in real time applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	1	-	-	-	-	-	-	1	2	-	-
CO-2	2	2	1	1	-	-	-	-	-	-	1	2	-	-
CO-3	2	1	1	1	-	1	-	-	-	-	1	2	-	-
CO-4	2	1	1	1	-	-	-	-	-	-	1	2	-	1

UNIT-I:

Introduction to Electrical Energy and DC Circuits: The role of Electrical Energy in modern life and various engineering branches- Circuit Concept – Types of Elements R-L-C parameters – Voltage and Current sources – Independent and dependent sources- Kirchhoff's laws – network reduction techniques: series, parallel, series parallel, star/delta transformations, Superposition theorem

UNIT-II:

Steady State AC Circuits: Representation of sinusoidal waveforms, average and RMS values, form factor and peak factor, phasor representation, Analysis of single-phase AC circuits consisting of R, L, C, series RL, RC, RLC combinations, real power, reactive power, apparent power, power factor - Three-phase balanced circuits, voltage and current relations in star and delta connections (Derivation only).

UNIT-III:

Electrical Machines: Transformer principle, types, Equivalent circuit, Regulation and Efficiency, DC generator principle, types Emf equation, DC motor principle, types, Back emf, Speed control of shun motor. Three phase induction motor, types, principle, torque Slip characteristics, working principle of Synchronous generator, principle of operation of synchronous motor.

UNIT-IV:**Electrical Installations:**

Components of LT Switch gear: SFU, MCB, ELCR, MCCB. Types of wiring, Need of earthing and types. Batteries-types-working and electrical characteristics of Lead-Acid battery, Elementary energy consumption calculations, Block diagram of UPS and its operation

UNIT-V:

Diode & Bipolar Junction Transistor: PN Junction Diode, Operation and Symbol, V-I Characteristics, Half Wave Rectifier, Full-Wave Rectifier, (Centre Tap and Bridge Rectifiers), Zener Diode, V-I Characteristics and applications, BJT, Construction (NPN and PNP transistors), Transistor configurations (CC, CB and CE), Transistor biasing, Transistor as an Amplifier.

TEXT BOOKS:

1. Basic Electrical Engineering, D. C. Kulshreshtha, 2nd Edition, TMH, 2019
2. Basic Electrical Engineering, P. Ramana, M. Suryakalavathi, G. T. Chandra Sekhar, 1st Edition, S. Chand, 2018
3. Electronic Devices and Circuits, S. Salivahanan and N. Suresh Kumar, 3rd Edition TMH, 2019

REFERENCES:

1. Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010
2. Electrical Engineering Fundamentals, Vincent Deltoro, 2nd Edition, Prentice Hall India, 1989
3. Electrical and Electronics Measurements and Instrumentation, A. K. Sawhney, 3rd Edition, Dhanpat Rai & Co., 1983
4. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, 3rd Edition, Tata McGraw-Hill, 2010
5. Engineering Circuit Analysis, William Hayt and Jack E. Kemmerly, 8th Edition, McGraw-Hill, 2013

ONLINE RESOURCE:

1. <https://nptel.ac.in/courses/117106108>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS2PH101) APPLIED PHYSICS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To explore the working principle of lasers and optical fibers through hands-on experiments
- To analyze the characteristics of semiconductors and devices
- To study resonance and magnetic induction in electromagnetic systems
- To synthesize the magnetic materials and study magnetic, dielectric properties of materials
- To develop skills in data analysis, interpretation, and scientific reporting

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Verify laser and optical fibre parameters through experimental study

CO-2: Characterize semiconductors and devices such as solar cell, laser diode etc

CO-3: Demonstrate resonance phenomena and realize tangent law of magnetism

CO-4: Understand the magnetic and dielectric properties of materials

CO-5: Apply scientific methods for accurate data collection, analysis, and technical report writing

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-2	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-3	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-4	3	3	1	2	1	2	1	3	2	-	1	1	-	-
CO-5	3	3	1	2	2	2	1	3	3	2	1	-	2	-

LIST OF EXPERIMENTS:

(Any 8 experiments are to be performed)

1.
 - a) Determination of wavelength of a laser using diffraction grating.
 - b) Study of V-I & L-I characteristics of a given laser diode.
2.
 - a) Determination of numerical aperture of a given optical fibre
 - b) Determination of bending losses of a given optical fibre.
3. Synthesis of magnetite (Fe₃O₄) powder using sol-gel method.
4. V-I characteristics of solar cell.
5. Magnetic field along the axis of current carrying coil – Stewart and Gee
6. Melde's Experiment.

7. Study of B-H curve of a ferro magnetic material.
8. Study of P-E loop of given ferroelectric crystal.
9. Determination of dielectric constant of a given material.
10. Determination of frequency of AC mains using sonometer.

TEXT BOOKS:

1. Engineering Physics Laboratory Manual/Observation, Physics Faculty of VNRVJIE
2. A Textbook of Practical Physics, S. Balasubramanian, M. N. Srinivasan, S Chand Publishers, 2017

REFERENCES:

1. Engineering Physics Lab Manual, Dr. Y. Aparna & Dr. K. Venkateswarao (V. G. S. Book links)
2. Engineering Physics Practicals, Dr. B. Srinivasa Rao, V. K. V. Krishna, K. S. Rudramamba, University Science Press, Imprint of Laxmi Publications, 2nd Edition

ONLINE RESOURCES:

1. <https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1>
2. <https://vlab.amrita.edu/index.php?sub=1&brch=280&sim=1518&cnt=1>

B. Tech. I Semester

(25HS2EN101) ENGLISH LANGUAGE AND COMMUNICATION SKILLS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

INTRODUCTION:

The aim of the English Language and Communication Skills Lab (ELCS Lab) is to develop the communicative skills of learners to function productively in academic and professional settings. This course focuses on developing Listening Skills, Reading Skills, Speaking Skills, Information transfer, Vocabulary, and pronunciation.

Professional students need reading strategies to explore scientific information and process the information. Further, they need to present and share this information in speaking and/or writing with their peers and scientific community. In addition to these, they also need good vocabulary for the presentation of their ideas. Practice is given not only in processing information to identify main ideas and relevant facts, to compare and contrast information and finally draw conclusions but also in presenting the same in speaking and writing.

COURSE OBJECTIVES:

- To train students to use neutral accent through phonetic sounds, symbols, stress and intonation
- To provide practice in vocabulary usage & grammatical construction
- To provide ample practice in LSRW skills and train the students in oral presentations, public speaking, role play, and situational dialogue
- To provide practice in defining technical terms and describing processes
- To equip students with excellent writing skills and information transfer skills

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Speak fluently with a neutral accent

CO-2: Use contextually apt vocabulary and sentence structures

CO-3: Make Presentations with great confidence

CO-4: Define technical terms and describe processes

CO-5: Write accurately, coherently, and lucidly

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	1	2	2	2	-	-	-	2	3	-	2	-	-	-
CO-2	1	2	2	2	3	3	2	2	3	2	2	1	-	-
CO-3	-	1	2	1	-	1	2	2	3	2	2	-	-	1
CO-4	2	2	2	2	2	-	-	1	3	2	2	2	2	2
CO-5	1	2	1	2	2	2	2	1	3	3	3	1	1	2

LIST OF EXPERIMENTS:

UNIT-I:

CALL

1. Common errors with Speech Sounds – Identify elements of Mother Tongue Influence,
2. Listening Skills – Types of Listening- Barriers – Active Listening Practice
3. Reading Skills -Skimming & Scanning-Reading for Gist -Reading for Specific Details

Oral Communication Lab:

1. Non-Verbal Communication
2. Formal and Informal English, Small Talk
3. Greetings – Introducing oneself & others - Elevator Pitch

UNIT-II:

CALL

1. **Activity:** Listening for information
2. Listening comprehension
3. **Reading Skills:** Reading comprehension-contextual vocabulary

Oral Communication Lab

1. Participatory Conversation
2. Role Playing in Formal Situations, Situational Dialogues – Two-Way (E. g. Elevator Pitch)
3. Media Etiquette, Conference Etiquette

UNIT-III:

CALL

1. **Phonetics:** Errors in Pronunciation, Neutralizing Mother Tongue Influence
2. **Listening Skills:** Listening for Gist
3. **Reading Skills:** SQ3R Method, Inferring

Oral Communication Lab

1. Describing Objects / Situations/ Places, People, Events,
2. Process Description

UNIT-IV:

CALL

1. **Listening Skills:** Techniques for Listening –Listening for Specific Details
2. **Reading Skills:** Making inferences

Oral Communication Lab

1. Story Telling; Story STAR, Sequencing, Creativity
2. Telling and Retelling Stories- Collage

UNIT-V:

CALL

1. **Listening Skills:** Listening for Evaluation, Writing the Summary
2. **Prompt Engineering:** Generating Appropriate Prompts, Keyword Extraction
3. Definition of a Technical Term; System Flow

Oral Communication Lab

1. Report Presentation
2. Short Presentation

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25PC2CE101) ENGINEERING GEOLOGY AND BUILDING MATERIALS TESTING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To identify the minerals by studying their physical properties
- To classify various types of rocks
- To determine the physical properties of building materials as per IS Code and understand their applications in civil engineering
- To evaluate the mechanical properties of building materials as per IS Code and understand their applications in civil engineering

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Describe and identify various minerals

CO-2: Distinguish and identify the rocks

CO-3: Evaluate the physical properties of building materials and assess their compliance with IS codes

CO-4: Test and verify the mechanical properties of building materials to ensure compliance with IS codes

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	-	-	2	3	3	-	1	2	-	-	1	3	-
CO-2	3	-	-	2	3	3	-	1	2	-	1	1	3	-
CO-3	3	2	-	-	1	-	-	1	2	-	2	1	1	3
CO-4	3	2	-	-	1	-	-	1	2	-	2	1	1	3

LIST OF EXPERIMENTS:

- Identification of physical properties of rock forming minerals: Feldspar, Quartz, Augite, Olivine, Mica, Garnet
- Identification of physical properties of ore forming minerals: Pyrite, Hematite, Magnetite, Galena, Graphite, and Bauxite
- Identification of Igneous rocks: Granite, Dolerite, Basalt, Pegmatite, Charnockite.
- Identification of Sedimentary rocks: Sandstone, Shale, Limestone, Conglomerate.
- Identification of Metamorphic rocks: Gneiss, Schist, Quartzite, Marble and Slate.
- Study of Models pertaining to folds, faults, joints and unconformities.
- Fineness of cement by Blaive's air permeability apparatus and Specific gravity of cement

8. Specific gravity of fine aggregate and coarse aggregate
9. Water absorption, Efflorescence and Compressive strength of bricks
10. Fineness modulus of fine aggregate and Bulking of sand
11. Flakiness Index and Elongation Index of coarse aggregate
12. Tensile strength of mildsteel

TEXT BOOKS:

1. Textbook of Engineering Geology, N. Chenna Kesavulu, 3rd Edition, Trinity Press
2. Concrete Technology, M. S. Shetty, 8th Edition, S. Chand & Co.
3. Engineering and General Geology, Parbin Singh, Katson Educational Series

REFERENCES:

1. Concrete Technology, A. M. Neville and J. J. Brooks, Prentice Hall
2. IS 4031 (part—2) -1999, Reaffirmed 2013, Method of physical test for Hydrulicment (Part-2) – Determation of fineness by Blaine Air permeability Method (Second Revision), Burean of Indain Standards, New Delhi, 2021
3. IS: 269 - 2015, (Reaffirmed 2020), Ordinary Portland Cement - Specification (Sixth Revision), Bureau of Indian Standards, New Delhi, 2020
4. IS: 383 - 2016, Coarse and Fine Aggregate for Concrete - Specification (Third Revision), Bureau of Indian Standards, New Delhi, 2016
5. IS 3495(Part- 1)- 2019, Burnt clay bricks- methods of test (Part-1) detemantion of compressive tests (4th revision), Burean of Indain Standards, New Delhi, 2019
6. IS 1608 (Part 1) - 2022, Metallic materials- Tensile testing – Part-1 methods of tests at room temperature, Bureau of Indian Standards, New Delhi, 2022
7. Handbook of Material Testing, Indian Railways Institute of Civil Engineering, 2022

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES2ME101) ENGINEERING WORKSHOP

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To know the different popular manufacturing process
- To gain a good basic working knowledge required for the production of various engineering products
- To provide hands on experience about use of different engineering materials, tools, equipment and processes those are common in the engineering field
- To identify and use marking out tools, hand tools, measuring equipment and to work to prescribed tolerances

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand various types of manufacturing processes

CO-2: Fabricate/make components from wood and steels through hands on experience

CO-3: Understand different machining processes like turning, drilling, tapping, etc.

CO-4: Understand electrical and electronic components and their assembly

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	-	-	1	-	-	-	-	3	3	2	-	2	1	-
CO-2	-	-	1	-	-	-	-	3	3	2	-	2	1	-
CO-3	-	-	1	-	-	-	-	3	3	2	-	2	1	-
CO-4	-	-	1	-	-	-	-	3	3	2	-	1	1	-

LIST OF EXPERIMENTS:

1. TRADES FOR EXERCISES:

- Carpentry:** Cross lap joint, Mortise & tenon joint
- Fitting:** L-fitting, Square fitting
- Tin smithy:** Rectangular tray, Rectangular scoop
- Arc welding:** Lap joint, Butt joint
- House Wiring:** Single lamp connection, Staircase connection
- Black smithy:** Round to Square, U-Hook
- Machine shop:** Step turning on lathe, Drilling & tapping

2. TRADES FOR DEMONSTRATION:

Plumbing, 3D printing, Power tools in construction and wood working.

TEXT BOOKS:

1. Workshop Manual, P. Kannaiah and K. L. Narayana, 3rd Edition, Scitech, 2015
2. Elements of Workshop Technology Vol. 1 & 2, S. K. Hajra Choudhury, A. K. Hajra Choudhury and Nirjhar Roy, 13th Edition, Media Promoters & Publishers , 2010

REFERENCES:

1. Manufacturing Engineering and Technology, Serope Kalpakjian, Steven R. Schmid, 4th Edition, Pearson Education India Edition, 2002
2. Manufacturing Technology-I, S. Gowri, P. Hariharan and A. Suresh Babu, Pearson Education, 2008
3. Processes and Materials of Manufacture, Roy A. Lindberg, 4th Edition, Prentice Hall India, 1998
4. Manufacturing Technology Vol-1 & 2, P. N. Rao, Tata McGraw-Hill, 2017

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS1MT105) ORDINARY DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

(Common to CE, ME and AE)

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematical Knowledge at Pre-University level

COURSE OBJECTIVES:

- To learn methods of solving first order differential equations and learn about its applications to basic engineering problems.
- To learn methods of solving higher order differential equations with constant coefficients and learn about its applications to basic engineering problems.
- To learn methods of solving higher order differential equations with variable coefficients and learn about its applications to basic engineering problems.
- To learn the physical quantities involved in engineering field related to vector valued functions
- To know the basic properties of vector valued functions and their applications to line, surface and volume integrals

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Formulate and solve the problems of first order differential equations

CO-2: Solve higher order differential equation with constant coefficients and apply the concept of differential equation to real world problems

CO-3: Solve the higher order differential equation with variable coefficients and apply the concept of differential equation to real world problems

CO-4: Find the gradient, divergence, curl and its physical interpretations

CO-5: Evaluate the line, surface and volume integrals and converting them from one to another

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	2	1	-	-	-	-	-	-	2	1	3
CO-2	3	3	1	2	1	--	-	-	-	-	-	2	1	3
CO-3	3	2	1	2	2	-	-	-	-	-	-	2	2	3
CO-4	3	2	2	2	1	-	-	-	-	-	-	1	2	3
CO-5	3	2	2	3	1	-	-	-	-	-	-	2	1	3

UNIT-I:

First Order Ordinary Differential Equations: Exact differential equations – Equations reducible to exact differential equations – linear and Bernoulli's equations – Orthogonal Trajectories (only in Cartesian Coordinates). Applications: Newton's law of cooling – Law of natural growth and decay.

UNIT-II:

Higher Order ODEs with Constant Coefficients: Higher order linear differential equations with constant coefficients: Non-Homogeneous terms of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x , $e^{ax} V(x)$ and $x V(x)$.

UNIT-III:

Higher Order ODEs with Variable Coefficients: Higher order linear Ordinary differential equations with variable coefficients, Method of variation of parameters, Euler-Cauchy ODEs, Legendre's ODEs.

UNIT-IV:

Vector Differentiation: Vector point functions and scalar point functions – Gradient – Directional derivatives -Divergence and Curl - Solenoidal and Irrotational vectors - Scalar potential functions – Vector Identities (without proof).

UNIT-V:

Vector Integration: Line, Surface and Volume Integrals. Theorems of Green, Gauss and Stokes (without proofs) and their applications.

TEXT BOOKS:

1. Higher Engineering Mathematics, B. S. Grewal, 36th Edition, Khanna Publishers, 2010
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, 5th Edition, Narosa Publications, 2016
3. Higher Engineering Mathematics, B. V. Ramana, 11th Reprint, Tata McGraw-Hill, 2010

REFERENCES:

1. Advanced Engineering Mathematics, Erwin Kreyszig, 9th Edition, John Wiley & Sons, 2006
2. Advanced Engineering Mathematics, Dennis G. Zill, 6th Edition
3. A Textbook of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications, 2008
4. Higher Engineering Mathematics, H. K. Dass and Er. Rajnish Verma, S. Chand

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS1CH102) ENGINEERING CHEMISTRY

(Common to CE, ME and AE)

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: General Chemistry and Basic Mathematics

COURSE OBJECTIVES:

- To analyze the quality of water for sustainable living
- To imbibe the conceptual knowledge of electrochemical processes and corrosion science
- To outline the importance of non-conventional energy sources and portable electric devices
- To acquire the knowledge about polymer science and its applications in various fields
- To recognize the significance of analytical techniques and materials in engineering, industrial, environmental, and biomedical applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Assess the specification of water regarding its usage in domestic & Industrial scenarios

CO-2: Interpret electrode potentials and predict the suitable corrosion control methods in safeguarding the structures

CO-3: Recognize the transformations in energy sources & battery technology

CO-4: Analyze the efficacy of polymers in diverse applications

CO-5: Identify the principles of UV-Visible and IR spectroscopy and materials in varied applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	2	2	2	1	1	1	1	2	1	-	1
CO-2	3	2	2	1	1	2	1	1	1	1	2	1	-	1
CO-3	3	2	2	1	1	1	1	1	1	1	2	1	-	1
CO-4	3	2	2	1	1	2	1	1	1	1	2	1	-	1
CO-5	3	2	1	1	1	1	1	1	1	1	2	1	-	1

UNIT-I:

Water and its Treatment: Introduction – Hardness, types, degree of hardness and units – numerical Problems. Estimation of hardness of water by complexometric method. Potable water and its specifications (WHO) – Steps involved in the treatment of potable water – screening, sedimentation, coagulation, filtration and disinfection by chlorination and break-point chlorination.

Boiler Troubles: Scales, Sludges, Internal treatment of boiler feed water – Calgon conditioning, Phosphate conditioning, Colloidal conditioning. External treatment methods – Softening of water by Ion-exchange process. Membrane separation –

Types of membranes [Microfiltration (MF), Ultrafiltration (UF), Nanofiltration (NF)], advantages and applications. Desalination of brackish water – Reverse osmosis.

UNIT-II:

Electrochemistry and Corrosion: Introduction – Electrochemical cell & galvanic cell, electrode potential, standard electrode potential, Nernst equation (no derivation). Types of electrodes – Reference electrodes – Construction, working principle, advantages, limitations and applications of Primary reference electrode (Standard Hydrogen Electrode – SHE), Secondary reference electrode (Calomel electrode) and Ion-selective electrode (Combined glass electrode).

Corrosion: Introduction – Definition, causes and effects of corrosion, Theories of corrosion – chemical and electrochemical theories of corrosion. Types of corrosion – galvanic, waterline and pitting corrosion. Factors affecting rate of corrosion – nature of metal (position, passivity, purity, areas of anode and cathode) & nature of environment (temperature, pH, humidity). Corrosion control methods – Cathodic protection Method – Sacrificial anode and impressed current cathodic protection methods.

UNIT-III:

Energy Sources and Battery Technology:

Fuels: Introduction and characteristics of good fuel, classification based on physical state.

Fossil Fuels: Petroleum – Refining of Crude oil, Cracking – Types of cracking – Moving bed catalytic cracking. Synthetic Fuels: Fischer-Tropsch process.

Gaseous Fuels: Composition and uses of LPG, CNG and Hythane. Green Hydrogen (generation by water electrolysis – PEM, Alkaline, Solid Oxide).

Batteries: Introduction – Classification of batteries – Primary, secondary, reserve batteries and fuel cells (differences and examples). Construction, working and applications of Zn-air (primary battery) and Lithium-ion battery (secondary battery), Characteristic features and Advantages of Metal ion batteries. (Sodium Ion Battery). Fuel Cells – Construction and applications of Proton Exchange Membrane Fuel Cell (PEMFC).

UNIT-IV:

Polymers: Definition – Classification of polymers – Types of polymerizations – Addition and condensation polymerization, differences between thermoplastics and thermosetting plastics. Plastics, Elastomers and Fibers: Synthesis, properties and applications of Teflon, Buna-S and Kevlar.

Fiber Reinforced Plastics (FRPs): Classification, features and applications.

Conducting Polymers: Definition, classification and applications.

Shape Memory Polymers: Definition, classification and applications.

Biodegradable Polymers: Introduction – Definition, Example: Features and applications of Polylactic acid (PLA).

UNIT-V:

Spectroscopic Applications and Engineering Materials: Interpretative spectroscopic applications of UV-Visible spectroscopy for analysis of pollutants in dye industry, IR spectroscopy in night vision – security, Pollution Under Control – CO sensor (Passive Infrared detection).

Self-healing Materials: Definition, working principle and applications (self-healing concrete).

Nanomaterials: Characterization by STM & AFM.

Lubricants: Definition and characteristics of a good lubricant – thin film mechanism of lubrication, properties of lubricants – viscosity, cloud and pour point, flash and fire point.

TEXT BOOKS:

1. Engineering Chemistry, P. C. Jain and M. Jain, Dhanpat Rai and Company, 2010
2. Engineering Chemistry, Rama Devi, Dr. P. Aparna and Rath, Cengage Learning, 2025
3. Engineering Chemistry, Thirumala Chary, Laxminarayana & Shashikala, Pearson Publications, 2020

REFERENCES:

1. Engineering Chemistry, Shashi Chawla, Dhanpat Rai and Company, 2011
2. Engineering Chemistry, Shikha Agarwal, Cambridge University Press, 2015
3. Textbook of Engineering Chemistry, Jaya Shree Anireddy, Wiley Publications
4. Engineering Analysis of Smart Material Systems, Donald J. Leo, Wiley, 2007
5. Challenges and Opportunities in Green Hydrogen, Paramvir Singh, Avinash Kumar Agarwal, Anupma Thakur, R. K. Sinha, Springer Nature, 2024

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25PC1CE102) SURVEYING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To understand the fundamental methods for linear and angular measurements
- To evaluate the reduced levels for the estimation of earthwork quantities, reservoir capacity and preparation of contours
- To implement trigonometric surveying, tacheometry, and curve setting for civil engineering applications
- To correlate the role of modern surveying and geomatics techniques in real-world engineering practice

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply basic surveying principles for accurate linear and angular measurements

CO-2: Develop contour map and estimate the earthwork quantities required for civil engineering projects

CO-3: Identify appropriate methods for curve setting and measurement of horizontal distances and vertical elevations

CO-4: Demonstrate the use and significance of modern surveying and geomatics instruments in engineering practice

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	2	2	-	-	-	1	-	-	-	-	-	3	-	-
CO-2	3	3	2	-	2	1	1	2	-	-	3	3	-	-
CO-3	3	3	2	2	-	1	-	2	-	-	3	-	-	2
CO-4	1	2	-	3	3	1	2	-	-	-	3	-	3	-

UNIT-I:

Introduction to Surveying: Classification; Principles of surveying; Plan, Map, Scale; Errors in surveying; Chain surveying; Principles and accessories.

Compass Surveying: Prismatic compass; angular measurements, bearings, dip, declination; local attraction.

UNIT-II:

Simple Levelling: Basic definitions; methods of levelling; Curvature and Refraction; Contour: contour interval, characteristics of contours, methods of plotting of contours, uses of contour maps

Areas and Volumes: Simpson's rule; boundaries with offsets at irregular intervals; coordinate method; level section; two level section; trapezoidal and prismoidal rule; capacity of a reservoir

UNIT-III:

Theodolite Survey & Traversing: Theodolite: parts, basic definitions, fundamental lines; measurement of a horizontal angle; repetition and reiteration; measurement of vertical angle

Trigonometric Levelling: Base of the object accessible, base of an inclined object accessible, reduced level of the elevated points with inaccessible bases.

UNIT-IV:

Tacheometric Surveying: Tacheometric measurement and constants; Inclined sight with staff held vertical; inclined sight with staff held normal to the inclined line of sight.

Curves: Elements of curve; designation of a curve; relationship between radius and degree of curve; types of curves; methods of setting out for simple circular curve: Linear methods - Offsets from long chord produced, Angular methods – Rankine's method of deflection angle.

UNIT-V:

Modern Field Survey Systems: Principle of Electronic Distance measurement; types of EDM instruments, total station, parts, accessories – advantages and applications, Global Positioning Systems-Segments, GNSS. Introduction, Basic concepts, perspective geometry of aerial photograph, relief and tilt displacements, flight planning.

TEXT BOOKS:

1. Surveying, Arora K. R., Vol. I, II and III, Standard Book House, 2019
2. Surveying, Duggal S. K., Vol. I and II, Tata McGraw-Hill, 2019
3. Surveying Vol. I & II, B. C. Punmia, Ashok Kumar Jain, Ashok Kr. Jain, Arun Kr. Jain., Laxmi Publications, 2016

REFERENCES:

1. Elementary Surveying - An Introduction to Geomatics, Charles D. Ghilani, Paul R. Wolf., Pearson India, 2021
2. Advanced Surveying: Total Station, GIS and Remote Sensing, Satheesh Gopi, Sathi Kumar R. and Madhu N., Pearson India, 2017
3. Surveying & Levelling, R. Subramanian, Oxford University Press, 2013

ONLINE RESOURCES:

1. <https://nptel.ac.in/courses/105104100>
2. https://onlinecourses.nptel.ac.in/noc22_ce78/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES1CE101) APPLIED MECHANICS

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Matrices and Calculus, Applied Physics

COURSE OBJECTIVES:

- To determine the resultant of coplanar concurrent and non-concurrent forces
- To state and apply the laws of friction
- To find centroids and moment of inertia
- To determine the member forces in pin jointed plane frames

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Analyze the systems using equilibrium conditions and apply the concepts of mechanics to engineering applications

CO-2: Apply the equations of equilibrium and analyze the bodies lying on rough frictional planes

CO-3: Determine the centroid and moment of inertia of built-up sections

CO-4: Analyze the pin jointed plane frames and tabulate the member forces

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	1	-	-	-	-	-	-	1	2	1	1
CO-2	3	3	2	1	-	-	-	-	-	-	1	2	1	1
CO-3	3	3	2	1	-	-	-	-	-	-	1	2	1	1
CO-4	3	3	2	1	-	-	-	-	-	-	1	2	1	1

UNIT-I:

Forces: Introduction to Engineering Mechanics – Basic concepts - Classification of a force system - Parallelogram law of forces - Triangle law of forces - Polygon law of forces – Law of transmissibility of forces – Principle of superposition – Lami's theorem - Free Body Diagram – Resultant – Equilibrant - Resultant of coplanar concurrent forces - Equations of Equilibrium of Coplanar Systems.

UNIT-II:

Moments: Moment of a force – Varignon's principle - Parallel forces - Resultant of parallel forces – Couple - Moment of a couple about any point lying in the plane - Resolution of a force into a force-couple and vice-versa - Resultant of coplanar non-concurrent forces – Beams – Degree of static indeterminacy - Determination of support reactions - Cantilever beams, Simply supported beams, Over hanging beams subjected to concentrated loads, moments, uniformly distributed loads and uniformly varying loads.

UNIT-III:

Friction: Types of Friction - Laws of friction - Equilibrium of bodies on rough horizontal and inclined planes - Equilibrium of connected bodies on rough horizontal planes, Ladder friction.

Centroid and Centre of Gravity: Introduction - Centroid - Centroids of lines, areas and volumes – Application to the centroids of built-up sections - Centre of gravity of bodies.

UNIT-IV:

Area Moment of Inertia: Introduction - Inertia - Inertia of areas - Radius of gyration - Polar moment of inertia - Parallel axis theorem – Perpendicular axis theorem - Moment of inertia of standard sections – Application to the moment of inertia of built-up sections.

UNIT-V:

Pin Jointed Plane Frames: Types of frames – Assumptions – Degree of static indeterminacy - Analysis of pin jointed frames by Method of joints, Method of sections.

TEXT BOOKS:

1. Engineering Mechanics, S. Timoshenko, D. H. Young, J. V. Rao and Sukumar Pati, 5th Edition, 2017, McGraw-Hill Publications
2. Engineering Mechanics: Statics and Dynamics, A. K. Tayal, 14th Edition, 2019, Umesh Publications
3. Engineering Mechanics, S. S. Bhavikatti, 8th Edition, 2021, New Age International Publishers

REFERENCES:

1. A Text Book of Engineering Mechanics, Dr. R. K. Bansal, 8th Edition, 2011, Laxmi Publications
2. Singers Engineering Mechanics: Statics and Dynamics, K. Vijaya Kumar Reddy & J. Suresh Kumar, 3rd Edition, 2024, B. S. Publications
3. Engineering Mechanics: Statics, R. C. Hibbeler, 15th Edition, 2021, Pearson Publications
4. Problems and Solutions in Engineering Mechanics, S. S. Bhavikatti & A. Vittal Hedge, 2nd Edition, 2016, New Age International Publishers

ONLINE RESOURCES:

1. <https://archive.nptel.ac.in/courses/112/106/112106286/>
2. <https://nptel.ac.in/courses/112103108>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25PC1CE103) BUILDING PLANNING AND CONSTRUCTION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Building Materials

COURSE OBJECTIVES:

- To comprehend the principles of building planning and discuss the different building components along with types of masonry and formwork
- To understand the various types of building services and their roles in functional and sustainable building design
- To identify and evaluate construction equipment to recommend the most suitable options for specific project needs and site conditions
- To interpret the features of green buildings and their importance in the present-day context of sustainable development

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Apply building principles, byelaws to plan a building and understand the various building components, formwork, and finishes used in construction

CO-2: Integrate building services into functional building designs in compliance with standards and sustainability principles

CO-3: Select and justify appropriate construction equipment for construction activities

CO-4: Analyze and explain the benefits, features, and rating systems of green buildings for promoting environmental sustainability

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	-	2	-	-	1	1	2	-	2	2	2	-	1
CO-2	3	-	2	-	-	2	-	3	-	-	2	2	-	-
CO-3	3	-	-	-	-	3	-	-	-	3	2	3	-	-
CO-4	3	-	-	-	-	1	1	-	3	2	2	-	-	1

UNIT-I:

Building Planning: Principles of Building Planning - Classification of Buildings - Building Bye Laws (NBC).

Building Components: Building Components- Structural and Non-Structural components - Foundations-Types and Applications - Damp Proof Course - Methods - Lintels – Types - Walls – Load Bearing and Non-Load Bearing Walls - Staircases – Types - Floors - Types - Roofs - Types.

UNIT-II:

Masonry: Stone Masonry – Types - Brick Masonry - Bonds - English and Flemish Bonds

Formwork and Finishing: Classification of Temporary Structures – Scaffolding Types, Finishes- Plastering-Pointing- Claddings – Types.

UNIT-III:

Building Services: Plumbing Services - Water Distribution System – Water Supply Fittings and Fixtures – Traps – Sanitary Appliances – Heating, Ventilation & Air Conditioning (HVAC) - Fire Protection – Fire Hazards.

UNIT-IV:

Construction Equipment: Trucks and other Earth Moving Equipment- Power Shovels – Dozers – Scrappers - Earth Compaction Equipment - Belt Conveyors - Concrete Mixers – Concrete Vibrators – Concrete Pumps – Bar Bending Machine - Cranes – Tower Cranes.

UNIT-V:

Introduction to Green Buildings: Definition of Green Buildings - Benefits of Green Buildings - Typical features of Green Buildings - Green Building Materials -Green Building rating systems: IGBC, BREAM, LEED, and GRIHA.

TEXT BOOKS:

1. Building Construction, B. C. Punmia, Ashok Kumar Jain, and Arun Kumar Jain, 12th Edition Laxmi Publications, 2025
2. Building Construction, P. C. Varghese, 2nd Edition, PHI Learning, 2017
3. Green Building Fundamentals, G. Harihara Iyer, 1st Edition, Notion Press, 2022

REFERENCES:

1. Fundamentals of Building Construction, T. D. Ahuja and G. S. Birdi, 4th Edition, Dhanpat Rai Publishing, 2012
2. Building Construction, S. C. Rangwala, 34th Edition, Charotar Publishing House, 2022

CODE BOOKS:

1. National Building Code of India, (Vol. I and II), 2016, Bureau of Indian Standards, New Delhi
2. Building Bye Laws, State and Central Governments, and Municipal Corporations

ONLINE RESOURCES:

1. <https://archive.nptel.ac.in/courses/105/103/105103206/>
2. <https://nptel.ac.in/courses/105102088>
3. <https://igbc.in/igbcgreenhomes/>
4. <https://www.grihaindia.org/griha-rating>
5. <https://www.gbig.org/collections/14555>
6. <https://breeam.com/>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES3CE101) COMPUTER AIDED ENGINEERING GRAPHICS

TEACHING SCHEME		
L	T/P	C
0	6	3

EVALUATION SCHEME				
D-D	SE	CP	SEE	TOTAL
10	20	10	60	100

COURSE OBJECTIVES:

- To acquire basic AutoCAD skills
- To understand various curves used in engineering practice, orthographic projections and projections of planes & solids
- To develop Sectional views of prisms, pyramids, cylinders and cones
- To visualize isometric projections, transformation of projections with application to brick bonds
- To study basic sign conventions of various building materials

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Construct various types of scales, various curves used in engineering practice using AutoCAD

CO-2: Sketch orthographic projections and sectional views of planes and solids

CO-3: Generate isometric projection and transformation of projections of Brick bonds

CO-4: Use sign conventions of building materials for line plan of building

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	1	1	-	-	3	-	-	1	-	-	1	3	-	-
CO-2	1	1	-	-	3	-	-	1	-	-	3	3	2	-
CO-3	1	1	-	-	3	-	-	1	-	-	3	3	2	-
CO-4	1	2	-	1	3	-	-	1	-	-	3	3	2	-

UNIT-I:

Introduction to Engineering Graphics: Principles of engineering graphics and their significance. – Size of drawing sheets, Types of lines, lettering, Dimensioning, Title block.

Introduction to Auto CAD: Views, Commands, and Printing Setup.

Scales & Conic Sections: Types of Scales–Plain Scale, Diagonal Scale. Conic Sections – Construction of Ellipse, Parabola, Hyperbola – General Method only.

UNIT-II:

Principles of Orthographic Projections: Principles of Orthographic Projections – Conventions – Projections of Points and Lines, Projections of Plane regular geometric figures – inclined to both planes.

UNIT-III:

Projections of Solids: Projections of Regular Solids – inclined to one and inclined to both the planes - Sectional views of Right Regular Solids – Prism, Cylinder, Pyramid, and Cone

UNIT-IV:

Isometric Projections: Principles of Isometric Projections, Isometric Views, Conventions, Isometric Projection of Planes - Application to brick bonds.

Transformation of Projections: Conversion of Isometric Views to Orthographic Views Conventions and Vice-versa.

UNIT-V:

Sign Conventions: Stone, Sand filling, Mortar, Concrete, Glass, Steel, Aluminium, Earth, Rock and Timber, doors, and windows.

Introduction to Line Plan: Development of Line plan of Building.

TEXT BOOKS:

1. Engineering Drawing, N. D. Bhatt, 55th Edition, Charotar Publishing House Pvt. Ltd., 2024
2. Engineering Drawing, Basant Agrawal, 3rd Edition, McGraw-Hill Education Pvt. Ltd., 2019

REFERENCES:

1. Engineering Drawing with AutoCAD, K. Venkata Reddy, 4th Edition, B. S. Publications, 2016
2. Civil Engineering Drawing-I, N. Sreenivasulu, S. Rama Rao, Revised Edition, Radiant Publishing House, 2017
3. Engineering Drawing, M. B. Shah, 2nd Edition, Pearson Education India, 2009
4. Engineering Drawing and Graphics, K. Venugopal, 4th Edition, New Age International, 2007

ONLINE RESOURCES:

1. <https://nptel.ac.in/courses/112103019>
2. <https://nptel.ac.in/courses/112105294>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS2CH102) ENGINEERING CHEMISTRY LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Basic Knowledge of Volumetric Analysis and Mathematics

COURSE OBJECTIVES:

- To understand the preparation of standard solutions and handling of instruments
- To determine and evaluate the water quality
- To measure physical properties like absorption of light, surface tension, pH, conductance, and viscosity of various liquids
- To conduct and collect the experimental data using different laboratory techniques
- To summarize the data and find the applicability to real world scenario

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Learn and apply the basic laboratory methodologies for the preparation of the standard solutions and handling of instruments

CO-2: Estimate the ions / metal ions present in domestic and industrial water

CO-3: Utilize the instrumental techniques to assess the physical properties of oils and water

CO-4: Analyze the experimental data to predict solutions for complex engineering problems

CO-5: Apply the skills gained to address societal issues related to real world scenario

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	3	2	1	1	-	1	1	1	1	1	2	-	-	-
CO-2	3	2	2	2	1	2	1	1	1	1	2	1	-	-
CO-3	3	2	1	2	1	2	1	1	1	1	2	1	-	1
CO-4	3	2	2	2	1	2	1	1	1	1	2	1	-	1
CO-5	3	2	2	2	1	2	1	1	1	1	2	1	-	1

LIST OF EXPERIMENTS:

VOLUMETRIC ANALYSIS:

1. Estimation of hardness of water by complexometric method using EDTA.
2. Determination of chloride content in the given sample water using Argentometric method.
3. Determination of Iron using Potassium Dichromate.

COLORIMETRY:

4. Estimation of copper present in the given solution by colorimetric method.

CONDUCTOMETRY:

5. Conductometric titration of Acid vs Base.
6. Conductometric titration of mixture of strong acid and weak acid by strong base.

pH METRY:

7. Titration of Acid vs Base using pH metric method.
8. Determination of viscosity of sample oil by Redwood Viscometer-I.
9. Estimation of acid value of given lubricant oil.
10. Determination of surface tension of a liquid by drop method using Stalagmometer.

PREPARATIONS:

11. Preparation of Bakelite.
12. Preparation of Nylon-6,6.
13. Determination of rate of corrosion of mild steel in the presence and absence of inhibitor.

14. VIRTUAL LAB EXPERIMENTS / CBP:

- Construction of advanced Fuel cell and it's working.
- Smart materials for Biomedical applications
- Batteries for electrical vehicles.
- Functioning of solar cell (materials used) and its applications.

TEXT BOOKS:

1. Laboratory Manual on Engineering Chemistry, S. K. Bhasin and Sudha Rani, Dhanpat Rai Publications
2. College Practical Chemistry, V. K. Ahluwalia, Sunitha Dhingra, Adargh Gulati, University Press
3. Practical Chemistry, Dr. O. P. Pandey, D. N. Bajpai, and Dr. S. Giri. S. Chand

REFERENCES:

1. Vogel's Textbook of Quantitative Chemical Analysis, G. N. Jeffery, J. Bassett, J. Mendham and R. C. Denny, Longmann, ELBS
2. Advanced Practical Physical Chemistry, J. D. Yadav, Goel Publishing House
3. Practical Physical Chemistry, B. D. Khosla, R. Chand and Sons
4. Lab Manual for Engineering Chemistry, B. Ramadevi and P. Aparna, S. Chand, 2022
5. Vogel's Textbook of Practical Organic Chemistry, 5th Edition

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25PC2CE102) SURVEYING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To apply the concepts of leveling for determining longitudinal and cross-sectional profile
- To develop contour maps through in-direct method of leveling
- To understand the principles of trigonometric and tacheometric surveying for measurement of horizontal and vertical distances
- To appreciate the applications of optical instruments for setting out of curve

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Establish benchmark and determine the reduced levels of various points

CO-2: Construct profile of a given ground location and develop contour maps

CO-3: Determine horizontal and vertical distances of accessible and inaccessible points

CO-4: Set out simple horizontal curves using optical instruments

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	1	1	-	-	1	2	-	3	2	-	-	3	-	-
CO-2	2	2	-	-	1	2	-	3	2	-	-	3	-	-
CO-3	2	3	-	2	1	2	-	3	2	-	-	3	-	-
CO-4	2	3	2	2	1	2	-	3	2	-	-	-	-	3

LIST OF EXERCISES:

1. Fly levelling
2. Differential levelling
3. Longitudinal levelling
4. Cross -sectional levelling
5. Contouring
6. Measurement of horizontal angle – Repetition method
7. Measurement of horizontal angles – Reiteration method
8. Measurement of vertical angles
9. Distance between two inaccessible points using the principles of trigonometric levelling
10. Determination of tacheometric constants
11. Distance between two inaccessible points using the principles of tacheometer
12. Setting out of simple circular curve

TEXT BOOKS:

1. Surveying, Arora K. R., Vol. I, II and III, Standard Book House, 2019
2. Surveying, Duggal S. K., Vol. I & II, Tata McGraw-Hill, 2019
3. Surveying Vol. I & II, B. C. Punmia, Ashok Kumar Jain, Arun Kumar Jain, Laxmi Publications, 2016

REFERENCES:

1. Elementary Surveying - An Introduction to Geomatics, Charles D. Ghilani, Paul R. Wolf., Pearson India, 2021
2. Advanced Surveying: Total Station, GIS and Remote Sensing, Satheesh Gopi, Sathi Kumar R. and Madhu N., Pearson India, 2017
3. Surveying & Levelling, R. Subramanian, Oxford University Press, New Delhi, 2013

ONLINE RESOURCES:

1. <https://nptel.ac.in/courses/105104100>
2. https://onlinecourses.nptel.ac.in/noc22_ce78/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25MN6HS102) ENVIRONMENTAL SCIENCE

TEACHING SCHEME		
L	T/P	C
0	6	3

EVALUATION SCHEME				
D-D	SE	CP	SEE	TOTAL
10	20	10	60	100

COURSE PRE-REQUISITES: Basic knowledge on environmental issues

COURSE OBJECTIVES:

- To recognize the principles of environment
- To identify the sources and causes of Environmental Pollution
- To analyze the causes of climate change and identify climate change mitigation and adaptation technologies
- To list out the benefits in creating circular economy and sustainability
- To emphasize the Environmental conventions, policies and regulations

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Gain knowledge about principles of Environment and allied problems

CO-2: Identify the causes and suggest remedial measures for Environmental Pollution

CO-3: Appraise the mitigation measures and adaptation technologies related to climate change

CO-4: Familiarize with the importance of circular economy towards sustainability

CO-5: Relate and implement environmental policies and regulations

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2	PSO-3
CO-1	2	2	1	-	1	2	2	1	1	1	2	1	-	1
CO-2	2	2	1	-	1	2	2	1	1	1	2	1	-	1
CO-3	2	2	1	-	1	2	2	1	1	1	2	-	-	-
CO-4	2	2	1	-	1	2	2	1	1	1	2	1	-	1
CO-5	-	1	1	-	-	2	1	1	1	1	-	-	-	-

MODULE 1:

Introduction to Environmental Science: Importance and Scope of Environmental Science. Overview of Environment & its components. Ecosystem function and uses. Biodiversity – Types and Values.

MODULE 2:

Synergy With Environment: Relationship between environmental science & society – Influence of Industry & infrastructure on environment (Specific case studies). Sources, Impacts and prevention of air, water and soil pollution.

MODULE 3:

Climate Change: Science behind climate change – factors responsible for climate change, Role of greenhouse gases – Global temperature rise & its impact on environment & human health. Carbon footprint – Briefing on Paris agreement, Identify

key sectors for low carbon footprint. Climate change mitigation & adaptation strategies.

MODULE 4:

Moving Towards Sustainability: EIA and EMP - importance, Sustainable agriculture – Importance of Organic farming and hydroponics. Role of AI & IOT for efficient management of environmental issues – Health, air, water, and soil. Sustainable living practices – Principles and methods (Linear Economy vs circular economy). Green Buildings.

MODULE 5:

Policies and Regulations: International Conventions: Montreal Protocol, COP15, Kyoto protocol (Salient Features). National Environmental Policies – Salient features of EPA act, Air act, Water Act, Forest and Wildlife act. National Green Tribunal Act (2010)- Features and Significance. Extended producer responsibility. Innovative approaches to waste management: Plastic, tyre and batteries.

TEXT BOOKS:

1. Living in the Environment, G. Tyler Miller, Scott E. Spoolman, 20th Edition, Cengage Learning
2. Environmental Studies, Rajagopalan, Oxford University Press, 2005
3. Introduction to Climate Change, Andreas Schmittner, Oregon State University, 2018. <https://open.oregonstate.edu/education/climatechange/>

REFERENCES:

1. Green Development: Environment and Sustainability in a Developing World, Bill Adams, 4th Edition, Routledge Publishers, 2021
2. Fixing Climate, Robert Kunzig & Wallace S. Broecker, Profile Books Publisher, 2009
3. Plastic Waste and Recycling-Environmental Impact, Societal Issues, Prevention and Solutions, 1st Edition, edited by Trevor Letcher, Academic Press, 2020
4. Waste Management Practices: Municipal, Hazardous, and Industrial, John Pichtel, CRC Press, 2020
5. Environmental Science: A Global Concern, William P. Cunningham, Mary Ann Cunningham, McGraw-Hill Education, 2022

ONLINE RESOURCE:

1. https://www.google.co.in/books/edition/Waste_Management_Practices/pG3SBQAAQBAJ?hl=en&gbpv=1&pg=PA1&printsec=frontcover